

Unit 7 Quest Study Guide DRAFT

Quadratic Functions | Draft New Format

Name: _____ Period: _____ Due: _____

How to use this guide: This draft keeps Unit 7 as a quest-style review: key features of parabolas, factoring, square-root property, quadratic formula, and graphing from a table. Each skill has a Version A and Version B so students can try a second similar problem after getting help.

Multiple-Choice Practice

1A. Y-Intercepts | Version A

What is the y-intercept of $f(x) = 2x^2 - 3x + 5$?

- A. (0, 2)
- B. (0, -3)
- C. (0, 5)
- D. (5, 0)

Work / thinking

1B. Y-Intercepts | Version B

What is the y-intercept of $g(x) = -x^2 + 4x - 7$?

- A. (0, -7)
- B. (0, 4)
- C. (-7, 0)
- D. (0, 1)

Work / thinking

2A. Axis of Symmetry | Version A

What is the axis of symmetry of $f(x) = x^2 - 6x + 8$?

- A. $x = -3$
- B. $x = 3$
- C. $x = 6$
- D. $x = -6$

Work / thinking

2B. Axis of Symmetry | Version B

What is the axis of symmetry of $f(x) = x^2 + 10x + 7$?

- A. $x = -5$
- B. $x = 5$
- C. $x = 10$
- D. $x = -10$

Work / thinking

3A. Vertex | Version A

What is the vertex of $f(x) = (x - 2)^2 - 9$? A. (2, -9) B. (-2, -9) C. (2, 9) D. (-9, 2)	Work / thinking _____ _____ _____
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3B. Vertex | Version B

What is the vertex of $f(x) = (x + 4)^2 + 1$? A. (4, 1) B. (-4, 1) C. (1, -4) D. (-4, -1)	Work / thinking _____ _____ _____
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4A. Factoring Roots | Version A

Solve $x^2 + 9x + 20 = 0$. A. $x = 4, 5$ B. $x = -4, -5$ C. $x = -4, 5$ D. $x = 0, -9$	Work / thinking _____ _____ _____
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4B. Factoring Roots | Version B

Solve $x^2 - 8x + 15 = 0$. A. $x = 3, 5$ B. $x = -3, -5$ C. $x = -3, 5$ D. $x = 0, 8$	Work / thinking _____ _____ _____
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5A. Square Root Property | Version A

Solve $x^2 = 49$. A. $x = 7$ B. $x = -7$ C. $x = -7$ or 7 D. $x = 24.5$	Work / thinking
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5B. Square Root Property | Version B

Solve $(x - 3)^2 = 25$. A. $x = -2$ or 8 B. $x = 2$ or -8 C. $x = 5$ D. $x = 28$	Work / thinking
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6A. Opening Direction | Version A

Which function opens downward? A. $y = x^2 + 3$ B. $y = 2x^2 - 5$ C. $y = -x^2 + 4x$ D. $y = 1/2x^2$	Work / thinking
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6B. Opening Direction | Version B

Which function opens upward? A. $y = -2x^2 + 1$ B. $y = -x^2 - 5$ C. $y = 3x^2 + x - 4$ D. $y = -1/2x^2$	Work / thinking
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7A. X-Intercepts | Version A

What are the x-intercepts of $y = (x - 1)(x + 6)$?	Work / thinking
A. (1, 0), (-6, 0)	_____
B. (-1, 0), (6, 0)	_____
C. (0, 1), (0, -6)	_____
D. no x-intercepts	

7B. X-Intercepts | Version B

What are the x-intercepts of $y = (x + 2)(x - 7)$?	Work / thinking
A. (-2, 0), (7, 0)	_____
B. (2, 0), (-7, 0)	_____
C. (0, -2), (0, 7)	_____
D. no x-intercepts	

8A. Quadratic Formula Setup | Version A

For $x^2 + 5x - 6 = 0$, what are a, b, and c?	Work / thinking
A. $a = 1$, $b = 5$, $c = -6$	_____
B. $a = 5$, $b = 1$, $c = -6$	_____
C. $a = 1$, $b = -5$, $c = -6$	_____
D. $a = -6$, $b = 5$, $c = 1$	

8B. Quadratic Formula Setup | Version B

For $2x^2 - 3x - 9 = 0$, what are a, b, and c?	Work / thinking
A. $a = 2$, $b = 3$, $c = -9$	_____
B. $a = 2$, $b = -3$, $c = -9$	_____
C. $a = -3$, $b = 2$, $c = -9$	_____
D. $a = -9$, $b = -3$, $c = 2$	

Free-Response Practice

Show clear work. Use the Version B problem as a second check after you have tried Version A.

9-A. Solving by Factoring | Version A

Solve $x^2 - x - 12 = 0$ by factoring. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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9-B. Solving by Factoring | Version B

Solve $x^2 + 2x - 35 = 0$ by factoring. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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10-A. Square Root Property | Version A

Solve $(x + 5)^2 = 64$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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10-B. Square Root Property | Version B

Solve $(x - 4)^2 = 81$. Answer: _____	Work / thinking _____ _____ _____ _____ _____
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11-A. Quadratic Formula | Version A

Solve $x^2 + 4x - 5 = 0$ using the quadratic formula. Answer: _____	Work / thinking
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11-B. Quadratic Formula | Version B

Solve $x^2 - 6x + 5 = 0$ using the quadratic formula. Answer: _____	Work / thinking
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12-A. Graphing Quadratics | Version A

For $f(x) = x^2 - 4x + 1$: a. Find the axis of symmetry. b. Find the vertex. c. State whether it opens up or down. Answer: _____	Work / thinking
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12-B. Graphing Quadratics | Version B

For $f(x) = -x^2 - 2x + 3$: a. Find the axis of symmetry. b. Find the vertex. c. State whether it opens up or down. Answer: _____	Work / thinking
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Student Self-Check Answer Bank

Important: This is not a worked solution key. It is a self-check bank. If your answer is not here, go back and check the setup, notation, and calculation.

Multiple-Choice Practice

Solve first, then check. Cross off each answer only after your work supports it.

A. $(-2, 0), (7, 0)$	A. $a = 1, b = 5, c = -6$
A. $(2, -9)$	A. $x = -5$
B. $x = 3$	B. $x = -4, -5$
C. $y = -x^2 + 4x$	A. $(0, -7)$
B. $(-4, 1)$	C. $(0, 5)$
C. $y = 3x^2 + x - 4$	B. $a = 2, b = -3, c = -9$
A. $(1, 0), (-6, 0)$	A. $x = 3, 5$
C. $x = -7$ or 7	A. $x = -2$ or 8

Free-Response Practice

Solve first, then check. Cross off each answer only after your work supports it.

$x = -5$ or $x = 13$	$x = 1$ or $x = 5$
a. $x = -1$; b. $(-1, 4)$; c. opens down	$x = -3$ or $x = 4$
$x = 1$ or $x = -5$	$x = -13$ or $x = 3$
$x = -7$ or $x = 5$	a. $x = 2$; b. $(2, -3)$; c. opens up

Final Check Before the Assessment

- ☐ I can identify y-intercepts, vertices, and axes of symmetry.
- ☐ I can solve quadratic equations by factoring.
- ☐ I can solve quadratic equations using square roots.
- ☐ I can use the quadratic formula.
- ☐ I can graph a quadratic and state its key features.